

# Ehsan Sharafian M.

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## PROFESSIONAL SUMMARY

Mechanical Engineering PhD candidate specializing in intelligent wearable systems for human movement analysis and digital health. Experienced in building end-to-end solutions that integrate IMU-based sensing, embedded hardware, smartphone applications, machine learning, and haptic feedback for gait analysis and rehabilitation. Skilled in Python, MATLAB, Android Studio/Java, and PyTorch, with six peer-reviewed journal publications.

## EDUCATION

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|-------------------------------------------------------------------------------------------------|-------------------|
| <b>Ph.D. in Mechanical Engineering</b>   University of Maine, Orono, ME                         | Expected Dec 2027 |
| GPA: 4.00/4.00   Dissertation: Smart Real-Time Gait Training System                             |                   |
| <b>M.S. in Mechanical Engineering</b>   Amirkabir University of Technology (Tehran Polytechnic) | Feb 2020          |
| GPA: 3.65/4.00   Thesis: Trajectory optimization of a robot arm in joint and Cartesian spaces   |                   |
| <b>B.S. in Mechanical Engineering</b>   University of Tehran                                    | Jul 2017          |

## TECHNICAL SKILLS

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|---------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Programming &amp; Data Analysis</b>      | Python, Java, C/C++, NumPy, Pandas, scikit-learn, PyTorch, SPSS                         |
| <b>Mobile &amp; Software</b>                | MATLAB, Simulink, TwinCAT, Android Studio, Unity, Git                                   |
| <b>Wearable &amp; Embedded Systems</b>      | IMU sensors, Arduino IDE, SparkFun modules, vibrotactile motors, PCB modules            |
| <b>Biomechanics &amp; Signal Processing</b> | Gait analysis, human activity recognition, sensor fusion, feature extraction, OpenSim   |
| <b>Robotics</b>                             | Surgical robotics, parallel manipulators, inverse kinematics, dynamics, control systems |
| <b>CAD</b>                                  | SolidWorks                                                                              |

## RESEARCH & ENGINEERING EXPERIENCE

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| <b>Graduate Research Assistant</b>   Biorobotics and Biomechanics Lab, University of Maine<br>Orono, ME                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Sep 2023 - Present |
| <ul style="list-style-type: none"><li>Development of real-time wearable sensing systems and Android-based mobile applications for gait analysis, activity recognition, fall-risk assessment, and home-based rehabilitation.</li><li>Built a smartphone-deployed gait assessment system using a single foot-mounted IMU to reconstruct foot-height trajectories and classify terrains across level walking, stairs, and ramps, achieving <b>96% accuracy</b>, <b>&lt;1.1 cm error</b>, and <b>112 ms latency</b>.</li><li>Engineered a lightweight human activity recognition system using two IMUs, biomechanical feature extraction, and classifiers including SVM and NB, achieving <b>99% accuracy</b> for normal locomotion and <b>93% accuracy</b> for atypical stair-negotiation strategies in older adults.</li><li>Architected a home-based haptic feedback gait-training system integrating IMUs, vibrotactile motors, custom Android applications, Arduino/SparkFun modules, and PCB circuitry for adults aged 65+.</li><li>Demonstrated up to <b>30% improvement</b> in stride length and walking speed during a single-session haptic feedback study, with minimal cognitive demand.</li><li>Led IRB-approved human-subject validation studies with <b>85+ participants</b> across gait analysis, activity recognition, and haptic feedback protocols.</li><li>Developed real-time algorithms and data pipelines in Android Studio/Java, Python, and MATLAB, combining signal processing, sensor fusion, biomechanical modeling, and machine learning for mobile health applications.</li></ul> |                    |

- Developed PLC-based control logic for a telesurgical laparoscopic robotic platform using Beckhoff TwinCAT and IEC 61131-3 Structured Text, with MATLAB/Simulink validation of inverse kinematics and initialization workflows.
- Built a Python/Kivy graphical user interface on Raspberry Pi for surgeon-facing robot control, system guidance, warnings, and improved human-robot interaction during testing and demonstrations.
- Coordinated hands-on validation sessions with **6 experienced surgeons**, using training, troubleshooting, and usability feedback to refine surgeon-facing robotic system workflows.

- Designed and assembled robotics and mechatronics setups for student research projects, integrating hardware, CAD, MATLAB workflows, and technical documentation.
- Mentored 6 undergraduate students across 5 research teams, guiding equipment use, troubleshooting, documentation, and project execution.
- Coordinated technical preparation for the 9th International Conference on Robotics and Mechatronics, including lab demonstrations, equipment readiness, and presentation logistics.

SELECTED HONORS

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|-----------------------------------------------------------|---------------------|
| Flagship Fellowship Award, University of Maine            | Sep 2024 - Sep 2025 |
| Graduate Student Government Award for conference proposal | Mar 2025            |

LEADERSHIP & SERVICE

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|-------------------------------------------------------------------------------------|---------------------|
| <b>Graduate Senator</b> , Department of Mechanical Engineering, University of Maine | Sep 2025 - Present  |
| <b>President</b> , Iranian Graduate Student Association, University of Maine        | Sep 2024 - Apr 2026 |

PUBLICATIONS

- 6 peer-reviewed (5 first-author) journal articles** in wearable sensing, biomechanics, rehabilitation engineering, and robotics.
- **Sensors** (2026) - Foot-height estimation and locomotion classification using a foot-mounted IMU and smartphone.
  - **IEEE Access** (2025) - IMU-based activity recognition for older adults using minimal biomechanical features.
  - **IEEE TNSRE** (2025) - Real-time haptic feedback effects on gait performance and cognitive load.
  - **Journal of Biomechanics** (2025) - Thigh extension and leg coordination in young and older adult walkers.
  - **International Journal of Robotics and Automation** (2023) - Robot-arm trajectory optimization for production-line applications.
  - **Robotica** (2022) - Screw theory-based constraint wrench analysis for a Delta manipulator.